

How Fidelis Deception® Helps Defend Against AI-Accelerated Intrusions

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Key Takeaways

AI Attacks Are Getting Faster. Most Defenses Are Not Keeping Up.

Security teams who watch intrusion activity day in and day out will tell you something that does not make it into most reports: the most dangerous thing about modern attacks is not sophistication. It is speed combined with patience.

The World Economic Forum's Global Cybersecurity Outlook 2026, drawn from over 800 CISOs, CEOs, and security executives across 92 countries, found that 87% of respondents identified AI-related vulnerabilities as the fastest-growing cyber risk throughout 2025. 94% expect AI to be the single most significant force reshaping network security in the year ahead.¹ The IBM X-Force Threat Intelligence Index 2026 explains why: exploitation of public-facing applications jumped 44% year-over-year, and AI tooling is actively compressing attacker decision cycles during reconnaissance, privilege escalation, and lateral movement.²

This is not a theoretical risk. Most enterprise intrusion detection systems were not built for this threat landscape. Traditional security methods depend on recognizable patterns. Signature-based approaches need the threat to already exist in a database. Anomaly

detection needs the attacker to behave unusually. AI-assisted attackers are designed to avoid both: using valid credentials, native admin utilities, and legitimate SaaS integrations to blend into normal behavior while moving laterally through an environment.

AI Makes Attacks Faster. It Does Not Change What Attackers Still Need.

AI-assisted tooling makes specific intrusion phases faster and more precise.

Reconnaissance is more systematic. Credential enumeration is more targeted. Lateral movement decisions, which path to take and which assets to prioritize, are increasingly guided by behavioral analysis on observed network data rather than slow manual trial and error.

This is documented in real campaigns. In July 2025, Ukraine's national CERT (CERT-UA) disclosed LAMEHUG, the first publicly documented malware to integrate a large language model directly into its attack flow.³ Attributed with moderate confidence to APT28 (Fancy Bear), LAMEHUG used an LLM to dynamically generate system reconnaissance commands, including hardware enumeration, process listing, and network connection mapping, without any hardcoded instructions. [Signature-based detection](#) and traditional intrusion prevention systems were structurally blind to it because there was no static pattern to match.³

What AI does not change is the dependency structure that every intrusion still runs on. Whether the attacker uses AI tooling or not, they need to map the environment, discover workable credentials, understand what assets are present, and find a viable path to their target. Every step requires the attacker to read and trust environmental signals: network topology, directory structures, service availability, file artifacts, credential stores.

Sophisticated attackers, even those augmented by AI agents, still trust the environment they see. That is exactly what [Fidelis Deception](#)[®] exploits.



How Fidelis Deception® Disrupts the Attack Lifecycle

- Attacker gains initial access

Phishing or credential theft lands a low-privilege foothold. AI-assisted tools begin scanning and mapping the environment to plan lateral movement.

- Reconnaissance hits deception layer

Fidelis Deception® populates the network with decoy assets, fake AD accounts, and breadcrumbs that look identical to real infrastructure. The attacker's map is wrong from the start.

- Attacker pursues deceptive credentials

Poisoned breadcrumbs, fake memory credentials, and false AD entries appear as high-value targets during enumeration. The attacker interacts with them, believing they are real.

- Alert fires the moment the decoy is touched

No behavioral threshold. No baseline is required. Any interaction with a deceptive object generates a high-confidence alert. False positives are near zero by design.

High-confidence detection

- Security team observes attacker TTPs in real time

While the attacker believes they are navigating real infrastructure, the security team tracks their moves, learns their techniques, and builds intelligence for hardening. Real assets stay untouched.

Why Signature-Based and Anomaly Detection Break Against AI Intrusions

Signature-based methods

Anomaly detection

Deception asks a different question

Traditional IDS / IPS

Fidelis Deception®

A Real Attack Scenario: What This Looks Like in Practice

Attack scenario with Fidelis Deception® deployed

What Fidelis Deception® Actually Covers Across Your Environment

Reconnaissance and network mapping

Faster reconnaissance is one of the clearest advantages AI-assisted tooling provides. Systematic scanning, network topology mapping, and service enumeration all happen

faster and more completely than manual methods. In a deception-rich environment, that thoroughness becomes a liability.

Fidelis Deception[®] maps the actual environment and uses terrain analysis informed by asset risk profiling to place decoys where attacker movement paths are most likely to intersect them. Decoys span laptops, servers, routers, cameras, printers, IoT devices, operating systems, applications, ports, and services across both on-premises and cloud environments. The more methodically an attacker scans, the more deceptive data they collect.

Credential discovery and Active Directory deception

Lateral movement detection

Attacker TTP intelligence and ongoing improvement

What This Means for Day-to-Day SOC Operations

Most advanced detection tools make analysts work harder before they work smarter. They need tuning, generate alert volume that must be sorted, and often require specialized expertise to operate. Against faster intrusion timelines, that overhead creates response lag that attackers can exploit.

Fidelis Deception[®] is designed to work the opposite way. The platform uses machine learning to analyze the environment, assess asset risk, and automate the deployment and ongoing updating of decoys and breadcrumbs based on that analysis. Security teams do not need to manually configure individual deceptive objects. Because any deception-layer alert reflects actual interaction with a decoy rather than a statistical anomaly, analysts can move directly to investigation without spending time validating whether the alert is real.

Telemetry, analysis, threat hunting, and incident response actions are consolidated in a single console. During an active event, every tool-switch introduces delay and the risk of dropped context. Threat hunting against deception-layer activity is backed by forensic visibility into attacker movements, which shortens investigation timelines and improves post-incident analysis quality.

For organizations running the [Fidelis Elevate XDR platform](#), deception-layer data correlates with network, endpoint, and sandbox signals, giving security teams cross-domain visibility across cloud environments, on-premises infrastructure, and identity systems in a single operational picture.

The More Sophisticated the Attack, the Better Deception Works

There is a counterintuitive dynamic that does not get discussed enough. More sophisticated attackers, including those using AI-assisted tooling for systematic reconnaissance and thorough environment mapping, tend to interact with more of the environment, not less. They enumerate more. They evaluate more credential sources. They assess more paths before committing to lateral movement.

Every one of those interactions is a potential detection event in a deception-rich environment.

Defenders who rely entirely on catching attacker mistakes in the real environment are betting on adversary errors. Deception technology does not require errors. It creates structured exposure points that sit directly in the path of normal attacker methodology. The more thorough the attacker is, whether human-directed or AI-assisted, the more likely they are to surface in the detection layer.

Given where evolving cyber threats are heading, that structural advantage matters more than ever. Deception becomes more effective as attacks become more thorough, which is precisely the direction the threat data points.